INCH-POUND

MIL-DTL-55288E
19 November 2012
SUPERSEDING
MIL-DTL-55288D
7 February 2005

DETAIL SPECIFICATION

ANTENNA AS-1729/VRC

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

Inactive for new design
4 June 1999

1. SCOPE

1.1 Scope. This specification covers one type of center-fed whip antenna intended for use in a vehicular communications system operating over the frequency range of 30.00 to 76.00 MHz and designated as Antenna AS-1729/VRC (see 6.1 and 6.6). The antenna consists of the following units:

Antenna Element AT-1095/VRC (Upper Whip Section).
Antenna Element, AS-1730/VRC (Lower Whip Section).
Matching Unit-Base, Antenna MX-6707/VRC (Base Assembly).

1.2 Appendix. Appendix A (Calibration and Alignment Procedures for the Matching Unit Base Alignment Cavity) forms a part of this specification.

1.3 Antenna elements. Quality assurance requirements in this specification applicable to Antenna Elements AT-1095/VRC and AS-1730/VRC spare part acquisitions are listed under 4.14.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to Tubesamps@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-T-152	-	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment.
MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-DTL-14072	-	Finishes for Ground Based Electronic Equipment.
MIL-I-17563	-	Impregnants for Cast or Power Metal Components.
MIL-PRF-23377	-	Primer Coatings, Epoxy, High-Solids.
MIL-DTL-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting, Lead & Chromate Free.
MIL-DTL-53030	-	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts.
MIL-STD-252	-	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices.
MIL-STD-276	-	Impregnation of Porous Metal Castings and Powdered Metal Components.
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-13231	-	Marking Of Electronic Items.

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY COMMUNICATION - ELECTRONICS COMMAND (CECOM)

CECOM DWG GL-SM-323477	Gages for Antenna AS-1729/VRC
CECOM DWG SC-A-46439	List of Accessories for Package Tester
CECOM DWG SC-C-208747	Antenna Tie-Down Assembly
CECOM DWG SC-DL-415405	Antenna Element AT-1095/VRC
CECOM DWG SC-DL-542000	Antenna Element AS-1730/VRC
CECOM DWG SC-DL-542020	Matching Unit-Base, Antenna MX-6707/VRC
CECOM DWG SM-B-542023	Alignment Specification
CECOM DWG SM-B-542115	RF Connector Adapter (Whip Admittance)
CECOM DWG SM-C-542002	Antenna Element AS-1730/VRC
CECOM DWG SM-C-542059	
CECOM DWG SM-D-415406	Finish Requirements for AT-1095/VRC
CECOM DWG SM-D-542009	Strain Relief Loop for Mast Section Assembly
CECOM DWG SM-D-542010	Envelope, Mast, Finish - Note 7
CECOM DWG SM-D-542021	Matching Unit-Base, Antenna MX-6707/VRC
CECOM DWG SM-D-542042	Cable-Mount Assembly
CECOM DWG SM-D-542043	RF Cable Assembly
CECOM DWG SM-D-542060	
CECOM DWG SM-D-542066	Matching Unit Case, Antenna
CECOM DWG SM-D-542070	

(Copies are available online at http://www.monmouth.army.mil/cecom/lrc/specstd/ssar-hp.html or may be obtained from usarmy.apg.cerdec.mbx.standardization-crx@mail.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM-D3935 - Material Polycarbonate (PC) Unfilled and Reinforced.

(Copies are available online at http://www.astm.org or may be obtained from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

HEWLETT PACKARD

Hewlett Packard Journal Volume 15, No. 6.

(Copies are available online at http://www.hp.com or may be obtained from Hewlett-Packard Corporation, 1501 Page Mill Road, Palo Alto, CA 94304.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL)

NCSL Z540.3 - Calibration of Measuring and Test Equipment, Requirements for.

(Application for copies of NCSL publications should be addressed to National Conference of Standards Laboratories, 2995 Wilderness Place Suite 107, Boulder, CO 80301-5404 or online at http://www.ncsli.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction AS-1729/VRC. The antenna shall be constructed in accordance with the applicable drawings indicated below:

Unit	NSN	Drawing	Data List
Antenna, AS-1729/VRC Assembly	5985009859024	CECOM DWG SM-D-542010	MIL-DTL-55288
Antenna Element AT-1095/VRC	5820008562728	CECOM DWG SM-D-415406	CECOM DWG SC-DL-415405
Antenna Element AS-1730/VRC	5985009859022	CECOM DWG SM-C-542002	CECOM DWG SC-DL-542000
Matching Unit-Base, MX-6707/VRC	5820009061115	CECOM DWG SM-D-542021	CECOM DWG SC-DL-542020
Antenna Tie-Down Assembly	4020009086416	CECOM DWG SC-C-208747	

3.2 First article. The contractor shall furnish four (4) first article samples of Antenna AS-1729/VRC for approval if required in contract.

3.3 Parts, materials, and process. The contractor shall use all items specified in the Government furnished Technical Data Package (TDP) or in the Government model. The TDP shall serve as the Program Parts Selection List (PPSL).

3.4 Castings. Zinc-alloy, aluminum-alloy, and magnesium-alloy die castings shall be used only when approved by the Contracting Officer for the particular application.

3.4.1 Quality of die casting (see 4.4). Quality of die castings shall be as follows:

a. Surface condition shall be clean, dry, and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

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d. The casting after impregnation, shall be air tight, and no leakage shall be allowed when tested in accordance with 4.7.3 and 4.9.

e. Flatness shall be as specified in Government-furnished drawings.

f. Dimensions shall be as specified in the Government-furnished drawings.

3.4.2 Impregnation of aluminum castings. All exterior aluminum castings shall be impregnated in accordance with MIL-STD-276, using class 3 impregnation material in accordance with MIL-I-17563. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.5 Plastic material and parts. When not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application. Plastic material and parts shall conform to the following, except that plastic material used for arc resistance shall be limited to the type specified. During the selection of plastic materials, consideration shall be given to tropicalization requirements.

3.5.1 Thermosetting. Unless in hermetically sealed enclosures, the thermosetting plastic material shall be treated.

3.5.2 Thermoplastic. Plastic molding material used in the construction of SM-D-542070 shall be unfilled, virgin, polycarbonate molding compound in accordance with ASTM-D3935. These materials shall not be treated and shall be masked, if necessary, during any treatment of equipment which might result in degradation of the electrical and physical characteristics of the material. Only fungistatic compositions shall be used or compounds modified with fungicides as an ingredient and capable of passing the fungus test, method 508, of MIL-STD-810.

3.6 Marking.

3.6.1 General. Marking shall conform to MIL-STD-13231 (see 4.4). Front panel marking shall be group I as described in that specification.

3.6.2 Visibility. Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.6.3 Serial numbers. The Matching Unit-Base, MX6707/VRC shall be serial numbered.

3.6.4 National Stock Number (NSN). The Matching Unit-Base MX-6707/VRC, the Antenna Element AS-1730/VRC (Lower Whip Section) and the Antenna Element AT-1095/VRC (Upper Whip Section) shall be marked with their applicable National Stock Numbers.

3.7 Service conditions. The equipment shall meet the following service conditions.

3.7.1 Temperature (see 4.7.1).

a. Operating. Ambient temperature in the range of +160°F, to -60°F. (The 160°F temperature includes effect of sun load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time. The antenna shall comply with 3.8.3 except that power handling capability shall be performed for only 15 minutes and at one frequency in each sub-band.

b. Non-operating. Exposure in the range of +160°F, to -60°F; exposure at the higher temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at one time. When returned to room temperature, the antenna shall comply with 3.8.1 and 3.8.2.1 and 3.8.3 with the exception of “a.” above.

3.7.2 Relative humidity (see 4.7.2). Antenna shall meet the requirements of 3.8.1, 3.8.2, and 3.8.4 when tested in accordance with 4.7.2.

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3.7.3 Immersion (see 4.7.3). The equipment shall show no evidence of leakage after being immersed in three feet of water for two hours. The antenna shall meet the requirements of 3.8.4.3 after the immersion test.

3.7.4 Vibration.

3.7.4.1 Vibration; internal, Matching Unit-Base, Antenna MX-6707/VRC (see 4.7.4.1). When vibrated in accordance with 4.7.41, The amplitude of any part, subassembly or structural member of the antenna base assembly shall not exceed twice the amplitude of the vibration applied to the assembly at any frequency between 10 and 55 cycles per second.

3.7.4.2 Vibration, whip assembly (see 4.7.4.2). The whip assembly shall be capable of being vibrated at the natural frequency which produces the greatest amplitude for a period of 2 hours without electrical or mechanical failure.

3.7.5 Bounce and shock. The equipment shall be capable of meeting the requirements of table I.

TABLE I. Bounce and shock.

Requirement paragraph	Inspection paragraph	Performance after test
Bounce	4.7.5	Specified performance 1/ No physical damage.
Shock ballistic	4.7.6	Specified performance 1/ No physical damage.

1/ The equipment shall meet specified performance for the following requirements: Base assembly resonance (see 3.8.1); VSWR-free space (see 3.8.2.1).

3.7.6 Impact (see 4.7.7). The antenna shall be capable of meeting the requirements of 3.8.1, 3.8.2.1 and 3.8.4 after subjection to the test of 4.7.7. Physical damage, if any, shall be limited to minor surface abrasions at the impact point, no deeper than .005 inch (0.13 mm).

3.7.7 Flexibility (see 4.7.8). The antenna shall withstand the flexing tests of 4.7.8 without electrical or mechanical damage.

3.7.8 Fungus (see 4.7.9). The antenna, with matching unit case (SM-D-542066) open, shall provide no fungus nutrients in the material, coating or contaminant form and shall not support fungal growth when tested. After being tested, the antenna shall show no more than six unrelated spots, each no greater than .015 square inch in area, of microbial growth (see 6.10.3) as evidenced by growth colonization (which includes branching and sporulation on or within each cubic foot, or fraction thereof, of antenna volume. Isolated instances of partial tubular germination shall not be included in this evaluation.

3.7.9 Salt fog (see 4.7.10). Equipment and assemblies shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface and shall meet specified performance for the following requirements: Whip measurements (see 3.8.4) and base assembly resonance (see 3.8.1).

3.7.10 Rain (see 4.7.11). The Antenna Matching Unit Base MX-6707/VRC when exposed to wind driven rain and high pressure washing hoses in accordance with 4.7.11 shall show no evidence of leakage.

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3.8 Electrical requirements.

3.8.1 Base assembly resonance (see 4.8.2). The base assembly (less cable assembly SM-D-542042) when subjected to the test specified in 4.8.2 shall resonate within frequency limits specified in table II.

TABLE II. Base assembly resonance values.

Sub-band	Resonance frequency (MHz)
1	39.80 ± 0.40
2	39.75 ± 0.65
3	39.95 ± 0.45
4	40.10 ± 0.50
5	39.80 ± 0.80
6	39.60 ± 0.60
7	39.60 ± 0.60
8	40.30 ± 1.00
9	39.90 ± 1.40
10	39.75 ± 1.25

3.8.2 Voltage standing wave ratio (VSWR).

3.8.2.1 VSWR free space (see 4.8.1.2). The VSWR of Antenna AS-1729/VRC as measured in free space at the rf input connector of the base assembly shall meet the requirements in table III.

3.8.3 Power handling capability (see 4.8.3). Antenna AS-1729/VRC shall be capable of continuous operation with an input rf power of 70 watts for one hour, without any damage to the antenna or any of its components. During this power handling capability test, the reflected and forward power shall be monitored and readings taken at the start of the test and at 5-minute intervals. The reflected power shall not change more than +1.5 watts through out the test.

3.8.4 Whip measurements.

3.8.4.1 Top whip section (see 4.8.4.1). When tested in accordance with 4.8.4.1, there shall be no visible effects or evidence of dielectric breakdown due to an applied voltage of 15,000 V rms.

3.8.4.2 Lower whip section (see 4.8.4.2). The lower whip section, when subjected to the same test described in 4.8.4.1, shall show no evidence of dielectric breakdown. Also, when tested to 4.8.4.2, the lower whip section shall show impedance of 50 ohms ± 2 percent for the lower portion of the coaxial conductor, and 125 ohms ± 2 percent for the upper portion of the coaxial conductor.

3.8.4.3 Whip assembly (see 4.8.4.3). The whip assembly admittance shall be measured at 75 MHz. The susceptance (jB) shall be 3.1 ± 4.0 millimhos and the conductance (G) shall be 27.0 ± 5.0 millimhos.

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TABLE III. VSWR values.
Voltage Standing Wave Ratio (VSWR)

Sub-band	Frequency (MHz)	Maximum VSWR in each sub-band 1/
1	30.0	3.0
1	33.0	3.0
2	33.0	3.0
2	37.0	3.0
3	37.0	3.4
3	42.0	3.4
4	42.0	3.0
4	47.5	3.0
5	47.5	3.0
5	53.0	3.0
6	53.0	3.0
6	56.0	3.0
7	56.0	3.2
7	60.0	3.2
8	60.0	3.2
8	65.0	3.2
9	65.0	3.2
9	70.5	3.2
10	70.5	3.0
10	75.95	3.0

1/ When measured as specified herein, maximum VSWR values listed include tolerances of test equipment and load terminations used during measurements.

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3.8.5 Control cable voltage (see 4.8.5). The rotary solenoid switch operating voltages at the base assembly shall be as stated in table IV. The manual/automatic band switch shall operate satisfactorily and index correctly when the dc voltage present on the band switch pins is as stated in table IV. The input voltage is to be 24 V dc.

TABLE IV. Control cable voltage. 1/

Band segment	Freq. MHz	AN/VRC-12 band	Voltages at pin						
			N	A	D	E	F	H	J
1	30-33	1	24	24	24	0	0	0	0
2	33-37	1	24	24	0	24	0	0	0
3	37-42	1	24	24	0	0	24	0	0
4	42-47.5	1	24	24	0	0	0	24	0
5	47.5-53	1	24	24	0	0	0	0	24
6	53-56	2	24	0	24	0	0	0	0
7	56-60	2	24	0	0	24	0	0	0
8	60-65	2	24	0	0	0	24	0	0
9	65-70.5	2	24	0	0	0	0	24	0
10	70.5-76	2	24	0	0	0	0	0	24

1/ Pin C is Ground - all voltages dc.

3.9 Bounce preconditioning. The MX-6707/VRC shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.10 Air seal (see 4.9). When the antenna is tested as specified in 4.9, there shall be no decrease in vacuum.

3.11 X-ray (see 4.11). When the AS-1730/VRC is tested as specified in 4.11, the internal assembly and dimensions above the stainless steel ferrule shall be observed to determine that the strain relief loop at top of AS-1730/VRC is in accordance with CECOM DWG SM-D-542009, and that there is continuity at the 50 to 125 ohm junction.

3.12 Interchangeability (see 4.12). Like units, assemblies, subassemblies, and replaceable parts shall be interchangeable (see 4.12).

3.13 Finish (see 4.13). The Antenna AS-1729/VRC or the Matching Unit-Base, or the Antenna Elements (Upper Whip Section, Lower Whip Section) when ordered separately shall be finished with Chemical Agent Resistant Coating (CARC) in accordance with procedures specified in 4.13. The finishing requirements as shown on CECOM DWG SM-D-415406, notes 1.8, 7, 10, and CECOM DWG SM-D-542010, note 7, or elsewhere, are suspended and replaced by CARC.

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3.13.1 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of antenna components and of solder shall not exceed 97 percent, by mass. Tin shall be alloyed with a minimum 3 percent lead, by mass (see 6.14).

3.14 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.15 Workmanship (see 4.15). The equipment shall be manufactured and assembled in a workmanlike manner and the applicable following references:

a.

Subject	Reference
Finish, protective	MIL-DTL-14072
Marking	MIL-STD-13231
Fungus proofing treatment	MIL-T-152

b. In this specification:

3.4 Castings
3.5 Plastic materials and parts
3.6 Marking.

4. VERIFICATION

4.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the manufacturer. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with NCSL Z540.3.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging, see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Conformance inspection (see 4.5).

4.3 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the item listed in 4.4, and the inspection specified for group A, group B, and group C (see tables V, VI, and VII, respectively.) Order and quantity of equipment to be subjected to environmental testing shall be as specified in table V. Requirements for replenishment parts may differ from these requirements. See the acquisition documents for specific requirements.

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TABLE V. Order of environmental tests.

Test	Unit 1	Unit 2	Unit 3	Unit 4
Non-damaging tests				
Temperature (see 4.7.1)	1			
Immersion (see 4.7.3)		1		
Potentially damaging tests				
Impact (see 4.7.7)	2	2		
Flexing (spring) (see 4.7.8)	3			
Vibration (see 4.7.4)		3		1
Bounce (see 4.7.5)		4		
Shock, ballistic (see 4.7.6)		5		
Rain (see 4.7.11)		6		
Humidity (see 4.7.2)	4			
Fungus (see 4.7.9) 1/			1	
Salt fog (see 4.7.10) 1/				2

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before packaging:

Item	Paragraph
Part, materials and processes	3.3
Castings	3.4
Marking	3.6

4.5 Conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be subjected to bounce preconditioning (see 3.9) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of bounce precondition shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedure and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.7). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, group C, or first article inspection shall be preconditioned after final assembly (see 4.6).

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4.5.1 Group A inspection. Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table VI. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air seal (see 4.9) shall be last.

TABLE VI. Group A inspection.

Inspection	Requirement paragraph	Inspection paragraph
Visual and mechanical:		4.10
Antenna Element AT-1095/VRC Antenna Element AS-1730/VRC Matching Unit-Base, Antenna MX-6707/VRC Antenna AS-1729/VRC		
Electrical:		
Matching Unit-Base, Antenna MX6707/VRC		
Base Assembly Resonance	3.8.1	4.8.2
Control Cable Voltage	3.8.5	4.8.5
Antenna Whip Elements		
Top Whip Section	3.8.4.1	4.8.4.1
Lower Whip Section	3.8.4.2	4.8.4.2
Whip Assembly	3.8.4.3	4.8.4.3
Matching Unit-Base, Antenna MX-6707/VRC		
Air Seal	3.10	4.9

4.5.2 Group B inspection. Group B inspection shall be performed on lots that have passed group A inspection. This inspection shall consist of the inspections listed in table VII. Each lot shall be subjected to sampling inspections in accordance with table VIII. Lots in which samples exhibit any failures shall be screened for that failure prior to units within the lot being subjected to group C inspection.

4.5.2.1 Order of inspection within group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE VII. Group B inspection.

Inspection	Requirement paragraph	Inspection paragraph
Electrical		
Antenna Element AS-1730/VRC		
X-ray	3.11	4.11
Antenna AS-1729/VRC		
Immersion	3.7.3	4.7.3
Interchangeability	3.12	4.12

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4.5.3 Group C inspection. This inspection shall consist of the tests specified in table IX and shall be performed on sample units that have been subjected to and met group A and group B inspections. Sample units shall be selected in accordance with 4.5.3.1. Group C inspection shall be performed in any order satisfactory to the Government, except that the rain test (see 4.7.11), shall be performed after all dynamic tests.

TABLE VIII. Sampling plan for group B inspection.

Lot size	Sample size
1	100%
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	6
151 to 280	7
281 to 500	9
501 to 1,200	11
1,201 to 3,200	13
3,201 and over	15

1/ The acceptance number in all cases is zero.

TABLE IX. Group C inspection.

Inspection	Requirement Paragraph	Inspection paragraph
	Subgroup 1 (see 4.5.3.1.1)	
Impact	3.7.6	4.7.7
Rain	3.7.10	4.7.11
	Subgroup II (see 4.5.3.1.2)	
Temperature	3.7.1	4.7.1
Flexing	3.7.7	4.7.8
Vibration - whip assembly	3.7.4.2	4.7.4.2
Bounce	3.7.5	4.7.5
Power handling capability	3.8.3	4.8.3
	Subgroup III (see 4.5.3.1.3)	
Humidity	3.7.2	4.7.2
Shock, ballistic	3.7.5	4.7.6
	Subgroup IV (see 4.5.3.1.4)	
Fungus	3.7.8	4.7.9 1/
Salt fog	3.7.9	4.7.10 1/

1/ This test shall be performed on a separate sample.

4.5.3.1 Sampling for inspection of equipment. Units selected for each group C inspection shall be selected without regard to their quality in accordance with the following.

4.5.3.1.1 Subgroup I. For this subgroup, two units for the first 50 production units shall be selected for this first group C inspection. For subsequent group C inspections, two units from each successive 250 or two per month if less than 250 are produced monthly, shall be selected.

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4.5.3.1.2 Subgroup II. For this subgroup, two units from the first 50 production units shall be selected for this group C inspection. For subsequent group C inspections, two units from each successive 500, or two every two months, if less than 500 are produced during a two month period.

4.5.3.1.3 Subgroup III. For this subgroup, two units from the first 50 production units shall be selected for the first group C inspection. For subsequent group C inspections, two units shall be selected from each successive 600, or two every three months, if less than 600 are produced during this three months period.

4.5.3.1.4 Subgroup IV. For this subgroup, two units from the first 50 production units only shall be selected. Subsequently, 2 units shall be tested every six (6) months until completion of production.

4.5.3.2 Noncompliance. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (a) the process and material, as applicable, and (b) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative for evaluation. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections or the inspections which the sample failed, at the option of the Government) and groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C re-inspection results have shown that the corrective action was effective (see 6.4).

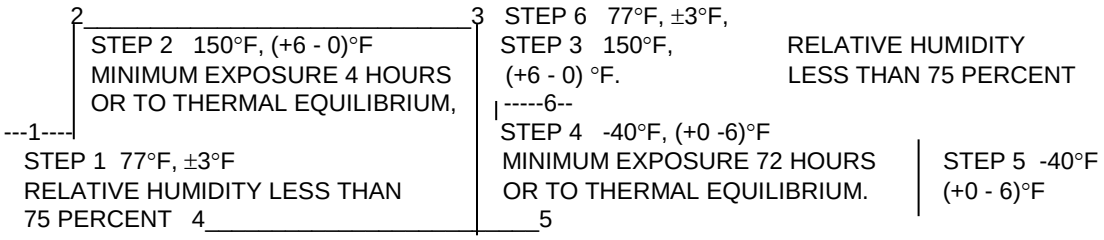
4.5.4 Re-inspection of conforming group C sample unit. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are re-subjected to and pass group A and group B inspections.

4.6 Bounce preconditioning. The antenna base assembly shall be placed in its normal operation position on the table of the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, NY, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a .031 inch (0.79 mm) strip of material under one corner or edge of the unit to a distance of 3 inches (76.2 mm) as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Service condition tests.

4.7.1 Temperature test (see 3.7.1). The Antenna AS-1729/VRC shall be subjected to the temperature cycle shown in MIL-STD-810 and in the temperature graph in 4.7.1.1.

4.7.1.1 Extreme temperature cycle.



a. Rate of temperature change: As rapidly as possible.
b. When measurements are taken on any step, practical thermal equilibrium shall first be obtained. Practical thermal equilibrium is attained when the temperature of any selected surface of the equipment changes less than 1°F during a half-hour period while the ambient temperature is steady.
c. Humidity is uncontrolled, except on steps 1 and 6,

4.7.1.2 Temperature tests (see 3.7.1). The following measurements shall be made on Antenna AS-1729/VRC before and after the temperature test cycle:

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Test	Paragraph
Free space VSWR	4.8.1.2
Power handling	4.8.3
Base assembly resonance test	4.8.2

The MX-6707/VRC shall be subjected to the test of 4.8.3 at 150°F and at -40°F to determine compliance with 3.8.3 except the 70 watt operation shall be for a 15 minute period at each temperature extreme and at one frequency in each sub-band, and a dummy load shall be used in lieu of the antenna test chamber.

4.7.2 Humidity test (see 3.7.2). The Antenna AS-1729/VRC shall be tested in accordance with method 507, induced procedure II of MIL-STD-810. Induced cycle 4 shall be used and the number of 24 hour cycles shall be 20. The following exceptions shall apply:

a. The equipment shall be removed from the humidity chamber for measurements which shall be made at ambient temperature and humidity.

b. The measurements shall be made as rapidly as possible after removal from the test chamber.

c. The seal on the antenna base assembly shall be broken for all five cycles by removal of the MX-6707/VRC matching unit case; however, the case shall also be subjected to the humidity cycles.

4.7.2.1 Performance. Measurements made and specified performance shall be as follows: The antenna shall meet full specification performance before, during, and after the test cycles, except voltage standard wave testing shall be performed only at steps 6 and 8.

Test	Requirement paragraph	Test paragraph
Base assembly antenna resonance	3.8.1	4.8.2
Free space VSWR	3.8.2.1	4.8.2.1
Whip antenna test	3.8.4	4.8.4

4.7.2.2 Failure. If any equipment fails to meet the performance specified in 4.7.2.1 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.7.2.1 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after condition, as specified in 4.7.2.1, it does not pass the test.

4.7.3 Immersion test (see 3.7.3). The antenna AS-1729/VRC shall be tested in accordance with method 512, procedure I, of MIL-STD-810.

4.7.4 Vibration.

4.7.4.1 Test for internal vibration (see 3.7.4.1). Internal vibration testing of the antenna base assembly shall be in accordance with the following procedure:

a. Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. To facilitate observation and measurement of amplitude of vibration, the equipment should be tested, provided it is secured to the vibration table in a manner similar to that used to mount the Antenna Base Assembly.

b. Vibrate the equipment successively in each of three mutually perpendicular directions over the frequency range of 10 to 55 cycles per second. The total excursion of the vibration shall be not less than .030 inch

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(0.76 mm). In each of the three perpendicular directions, change the frequency in discrete steps of one cycle per second and maintain each frequency for at least 10 seconds.

c. At the completion of the test, the base assembly resonance test (see 4.8.2), shall be performed to determine compliance with 3.8.1.

4.7.4.2 Vibration, whip assembly (see 3.7.4.2). The whip assembly shall be vibrated to determine its natural frequency exhibiting the highest vibration amplitude, and then the whip assembly shall be vibrated for two hours at this natural frequency. At the completion of the test, the whip antenna test of 3.8.4 shall be performed to determine compliance with 4.8.4. The total excursion of the applied vibration shall be not less than .030 inch (0.76 mm).

4.7.5 Bounce test (see 3.7.5). The Antenna AS-1729/VRC shall be tested on a package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, NY or equal. Accessories shall be selected from those listed on CECOM DWG SC-A-46439. The tests shall be as follows:

a. Secure the equipment to the 48 inch by 48 inch (1219 mm by 1219 mm) vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch (51 mm) breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches (178 mm) from the edge of the adapter plate.

b. Cover the tester bed with a panel of .500 inch (12.70 mm) plywood, with the grain parallel to the drive chain. Space sixpenny nails with heads below the surface, at 6-inch (152.4 mm) intervals around all four edges and at 3-inch (76 mm) intervals in a 6-inch (152 mm) square in the center.

c. Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches (76 mm) and not less than 1 inch (25 mm). Additional barriers may be used to safeguard personnel, provided that the fore and aft motion of the adapter plate against the backstop is not restrained.

d. Operate the package tester, shafts in phase, for a total of 3 hours at 283 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction.

e. During the final 15 minutes of bounce testing, the antenna is to be energized by a 1,000 cycle modulated rf carrier of minimum power in the frequency range from 30 to 75.95 MHz. A Government approved communication type receiver located a minimum of 100 feet from the antenna under test shall be tuned to the frequency of the AS-1729/VRC under test. The received signal shall be clear with no interruptions.

f. At the completion of three hours of bounce testing, the equipment shall be tested to 4.8.1.2 and 4.8.2 to demonstrate compliance with 3.8.1 and 3.8.2.1.

4.7.6 Shock test, ballistic (see 3.7.5). The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The antenna shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows. The sequence of the test blows is to be: (a) 1-foot, and 5-foot blows on the back; (b) 1-foot, 3-foot, and 5-foot blows on the top; and (c) 1-foot, 3-foot, and 5-foot blows on the side. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. At the completion of the 9 blows, the antenna shall be tested to 4.8.2 and 4.8.4 to demonstrate compliance with 3.8.1 and 3.8.4.

4.7.7 Impact (see 3.7.6). The Antenna AS-1729/VRC, secured on a vehicle in a normal operating position, shall be subjected to 25 repeated blows in any direction, against a 4 inch by 4 inch (101.6 mm by 101.6 mm) oak beam, when the vehicle is traveling at a rate of 25 miles per hour. The point of impact shall be at the mid-point of the whip assembly. The oak beam shall be stationary at the point of impact. After being subjected to this test, the antenna shall meet the whip antenna test of 4.8.4 and the x-ray test of 4.11.

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4.7.8 Flexing test (see 3.7.7). One base assembly spring shall be flexed 40,000 cycles with each cycle consisting of a deflection of the top of the spring +30° from the unflexed position, to -30°, and return to normal. The 40,000 cycles shall be accomplished by rotating and cycling the spring approximately 13,333 cycles at each of three (3) 120° positions. A second spring shall then be flexed 5,000 cycles, with one cycle consisting of a deflection of +90° from the unflexed position, to -90°, and return to normal. The 5,000 cycles shall be accomplished by rotating the spring approximately 1,666 cycles at each of three (3) 120° positions. The test shall be performed at a temperature of -40°F. The flexing rate shall be 10 ± 1 cycle per minute. Tests of springs shall be performed with rf cable SM-D-542043 assembled in it in holders SM-D-542060 and SMC-542059 or equivalent fixtures. The rf cable shall be tested for continuity before and after the flexing procedure.

4.7.9 Fungus (see 3.7.8). The antenna shall be tested in accordance with method 508, procedure I of MIL-STD-810. The antenna matching unit case (SM-D-542066) shall be opened, the AT-1095/VRC and AS-1730/VRC elements shall not be connected to each other or the MX-6707/VRC base, the parts and materials of the antenna internal assembly, the interior of the case, the interior surfaces of the cable mount assembly (SM-D-542042), as well as the exterior of the antenna shall be sprayed with the spore suspension. The antenna shall be subjected to the test in this open condition. There shall be abundant growth colonization (see 6.10) on 50 percent or more of the control items after 7 and 28 days. The inability of the antenna to meet the requirements of 3.7.8 shall constitute failure of this test.

4.7.10 Salt fog (see 3.7.9). The equipment and assemblies shall be exposed in their operating configuration to salt fog in accordance with method 509 of MIL-STD-810 for 48 hours. Equipment and assemblies shall not be opened or operated during the test cycle. Immediately before and after the test cycles, the equipment and assemblies shall be tested for compliance with the requirements specified in 3.7.9.

4.7.11 Rain test (see 3.7.10). The Antenna Matching Unit Base MX-6707 shall be subjected to the blowing rain and water-tightness tests of procedures I and II in method 506 of MIL-STD-810. The choice of related test conditions for each of the two tests are as follows:

Condition	Procedure I	Procedure II
Test Item Configuration	Operational, on tank and wheeled vehicles.	Washing with high pressure hoses on both
Rainfall Rate/Pressure	5 inches per hour	150 psi at 3 feet
Wind Velocity/GPM	40 mph	100
Duration	45 minutes	10 minutes
Temperature, Unit Base, °F	Rain temperature +18°F minimum at beginning.	Hose water temperature +18°F minimum at beginning.

4.8 Electrical tests.

4.8.1 Voltage standing wave ratio (VSWR).

4.8.1.1 Standard test condition. All electrical free space measurements, unless otherwise specified, shall be made in a reflection free environment. The test setup shall be located in an area that is away from any structure or geographical feature which would produce reflections and measurably alter the free space impedance. The antenna under test shall be elevated sufficiently to minimize the effects of terrain and all tests shall be performed over a ground plane consisting of a solid 10 foot by 10 foot copper sheet ground plane. Input impedance shall be referred to 50 ohms and the frequency of the source shall be accurate to within ±0.1 MHz. The magnitude of rf leakage in any of the test equipment plus any subassembly of the AS-1729/VRC Antenna shall be sufficiently low so that it cannot introduce errors into any electrical measurement.

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4.8.1.2 Free space VSWR test (see 3.8.2.1). The entire AS-1729/VRC Antenna with its permanent dust cover in place shall be mounted over a ground plane and under the conditions of 4.8.1.1 above. The VSWR shall be measured by any commercially standard means acceptable to the Government to determine compliance with 3.8.2.1.

4.8.1.3 Data and correlation records. Measurement procedures for all electrical tests made shall conform to the requirements of this specification. An up to date file describing equipment and procedures shall be maintained by the contractor. Included in the contractor's file, as a minimum, shall be:

a. Design and engineering drawings of all special equipment, including all electrical values and mechanical details and all production test procedures for piece parts, subassemblies, and the complete item on order.

b. A complete test layout for all electrical tests performed including complete details regarding type and length of cables used, type of connectors used, construction and adjustment test or positioning fixtures used, any wire and cable dress and routing, and any critical dimensions and adjustments that must be adhered to.

c. Complete step-by-step method of operation of the test equipment and methods of computation.

d. Calibration records on all test equipment used in taking measurements.

e. Records showing equivalency to devices specified contractually when an "or equal" device is permitted by the Government. All of the data listed above shall be available for review by Government personnel when requested. All disputes regarding the accuracy or adequacy of the contractor's records shall be referred to the acquiring activity for resolution.

4.8.2 Base assembly resonance test (see 3.8.1). The base assembly (less cable-mount assembly SM-D-542042) when tested in accordance with the following Base Assembly Resonance Test Procedure shall resonate within frequency limits given in 3.8.1.

Base assembly resonance test procedure:

a. Set up the equipment as shown on figure A-1 (see appendix A).

b. Insert MX-6707/VRC (less cable mount, assembly SM-D-542042) into Matching Unit-Base Alignment Cavity and secure with grounding pins.

c. Connect control cable from Alignment Cavity Control to connector J2 of MX-6707/VRC under test.

d. Turn on equipment and allow to warm up for a minimum of 30 minutes.

e. Set the Alignment Cavity Control sub-band selector switch to sub-band 1.

f. Depress 10 MHz sweep width switch on Sweep Generator.

g. Adjust the Sweep Generator variable sweep width control to maximum (initially).

h. Set the Sweep Generator sweep mode/time switch to .01 seconds.

i. Set the counter measurement interval switch to .01 seconds.

j. Adjust the Sweep Generator variable sweep width control for approximately 4 MHz display bandwidth on oscilloscope with applicable table II sub-band resonance frequency at center of oscilloscope. Adjust center frequency control as necessary.

k. Adjust Sweep Generator variable amplitude control for approximately full vertical display on oscilloscope.

l. Adjust the Sweep Generator dot position control until dot is displayed at the peak of the response curve. Read and record the frequency displayed on the counter.

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m. Adjust the Alignment Cavity Control sub-band selector switch to next sub-band.

n. Repeat step j, k, and l.

o. Repeat steps m and n until all sub-bands have been measured.

NOTES:

(1) The Matching Unit-Case (SM-D-542066) shall be on the MX-6707/VRC under test.

(2) For each Matching Unit-Base Alignment Cavity, the contractor shall establish and maintain a standard MX-6707/VRC in accordance with A.3.1.

(3) The contractor maintained standard MX-6707/VRC unit shall be used at the beginning of each work day, or whenever there are questionable readings to verify that the Matching Unit-Base Alignment Cavity is operating satisfactorily.

4.8.3 Power handling capability test (see 3.8.3). All power handling capability tests shall be performed using the antenna test chamber. Measurements shall be made at one frequency in each sub-band of table II for a total of ten (10) frequencies. The power handling test time for each frequency selected shall be one (1) hour. Each operating hour shall be followed by a non-operating one-half hour for purpose of cooling. Frequencies selected for these tests should exhibit approximately the same VSWR in the chamber as is required by table III.

4.8.4 Whip antenna test.

4.8.4.1 Top whip section test (see 3.8.4.1). The test shall be performed in accordance with MIL-STD-202, method 301, to determine compliance with 3.8.4.1.

a. The test voltage shall be 15,000 V rms.

b. The test voltage shall be applied along the full length of the upper whip section, and along the lower whip section from a point one inch above the stainless steel ferrule to a point an inch below the metallic captivating pin.

4.8.4.2 Lower whip section test (see 3.8.4.2). Measurements shall be made to determine the characteristic impedance of the 50 ohm and 125 ohm cables within the whip section using the "Time Domain Reflectometry" test method as outlined in Hewlett-Packard Journal, Volume 15, No. 6, dated February 1964.

4.8.4.3 Whip assembly test (see 3.8.4.3). For the purpose of this test, the whip assembly shall consist of:

a. Antenna Element, AT-1095/VRC, upper section.

b. Antenna Element, AT-1730/VRC, lower section.

c. The Base Spring.

This whip assembly shall be mounted on a 10-foot by 10-foot copper sheet (see figure 1). The admittance (normalized to 20 millimhos of the assembly shall be measured at the end of a 27 inch (685.8 mm) length of RG-58C/U coaxial cable connected to the "RF Connector-Adaptor, SM-B-542115", at 75.0 MHz. For admittance information, see appendix A.

4.8.5 Control cable voltage (see 3.8.5). The Matching Unit-Base, Antenna MX-6707/VRC antenna control connector J2, pin voltages, and manual/automatic band switch indexing shall be measured for compliance with table IV.

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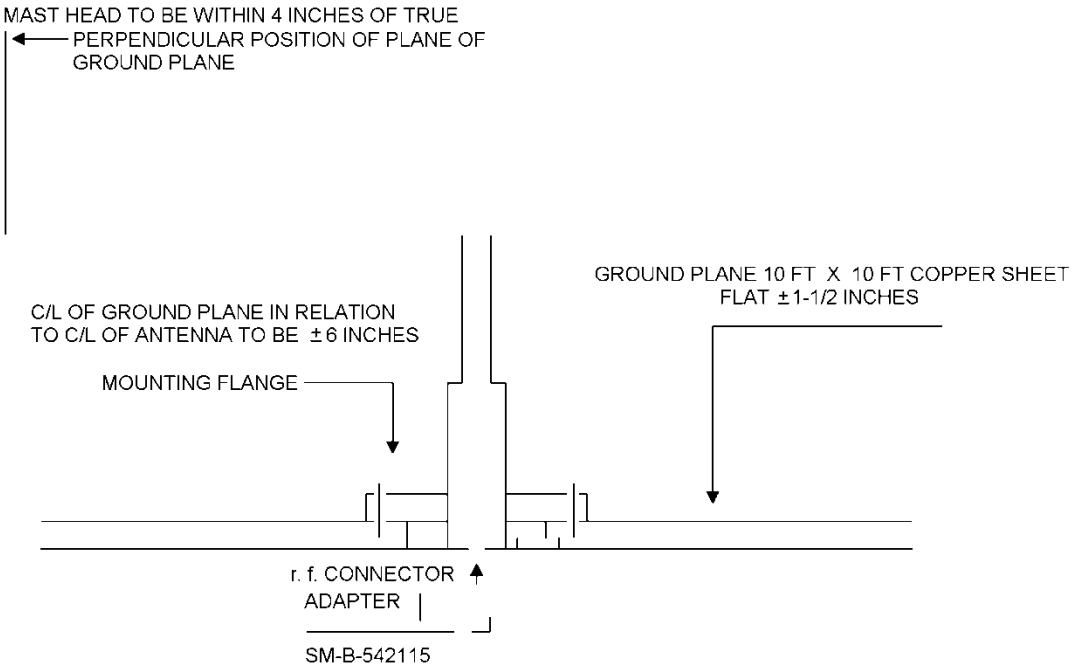


FIGURE 1. Antenna assembly under test
Ground plane and mast located in reflection free environment.

4.9 Air-seal test (see 3.10). The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment enclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 2 minutes. During this 2-minute period, there shall be no decrease in vacuum. The gauge used for measurement of vacuum shall be of such accuracy that any decrease in vacuum can be readily determined.

4.10 Visual and mechanical inspection. The equipment shall be examined for the defects listed in MIL-STD-252.

4.11 X-ray (see 3.11). The AS-1730/VRC Antenna Element shall be x-rayed along the entire area above the stainless steel ferrule to determine compliance with 3.11.

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4.12 Inspection of interchangeability (see 3.12). The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirements of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies or major units; and location of mating parts on the plug-in assembly or major unit.

d. Size and form of special threads.

e. Conformance of AT-1095/VRC and AS-1730/VRC to drawing CECOM DWG GL-SM-323477.

4.13 Finish and pretreatment (see 3.13).

a. Where technical drawings and the equipment specification specify olive drab finish for piece parts, subassemblies, and assemblies, these shall not be finished with the lusterless forest green paint conforming to film designation "V" in MIL-DTL-14072 “V”, but shall be finished with the Chemical Agent Resistant Coating (CARC).

b. Requirements are as follows:

(1) Finish: Finish pretreatment and finishes of external surfaces shall be as follows:

(a) For ferrous metals, primer is to be in accordance with MIL-DTL-53022 or MIL-DTL-53030. Thickness of primer .0010 to .0015 inch (0.025 to 0.038 mm).

(b) For non-ferrous metals, primer is to be in accordance with MIL-PRF-23377. Thickness of primer .0006 to .0009 inch (0.015 to 0.022 mm).

(c) In order to paint the Lexan Base with CARC, a primer is required. The USA Belvoir Research and Development Center recommends using primer MIL-DTL-53022. The color of the part as molded (prior to painting) shall be as close as possible to Green 383 by pre-mixing the material with color pigment.

(d) Final finish coating shall be type II Green 383. Final finish is applied at a minimum of .0018 inch (0.045 mm) dry in two coats.

(e) Antennas: Finish coat on signal radiating or receiving elements where specified on drawings, must be type II, Green 383.

(2) Color: Color of external surfaces shall be type II Green 383.

c. The equipment shall be marked with the word "CARC" in close proximity to the ID nameplate or N.S.N., stenciled in accordance with MIL-STD-13231, group II using black ink, one-half inch high, one-quarter inch high may be used on whip assembly.

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4.14 Quality assurance requirements applicable for antenna elements AT-1095/VRC and AS-1730/VRC:

Paragraph	Subject
4.1	Responsibility for compliance.
4.2	Classification of inspection.
4.4	Inspections covered by subsidiary documents.
4.5	Conformance inspection of equipment before packaging.
4.7	Service condition tests.
4.7.1	Temperature test: Non-operating procedure only. When returned to room temperature, the antenna element shall comply with test in 4.8.4.3 only.
4.7.2	Humidity test. An MX-6707/VRC shall not be included in the humidity chamber. Under 4.7.2.1, units shall be tested for performance in accordance with 4.8.4.3 only.
4.7.4.2	Vibration, whip assembly.
4.7.7	Impact.
4.7.9	Fungus.
4.7.10	Salt fog.
4.8	Electrical tests.
4.8.4	Whip antenna test.
4.8.4.1	Top whip section test.
4.8.4.3	Whip assembly test.
4.10	Visual and mechanical inspection.
4.11	X-ray.
4.12	Inspection of interchangeability.
4.13	Finish and pretreatment (CARC paint finish).
4.15	Workmanship.

4.15 Workmanship (see 3.15). The workmanship used in the manufacturing of Antenna AS-1729/VRC or portions thereof is to be in accordance with best manufacturing practices. Other military standards and military specifications listed in 3.15 apply as they are applicable to this antenna.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Antenna AS-1729/VRC is intended for use with and is automatically tuned by Radio Sets AN/VRC-12, AN/VRC-43 through AN-VRC-49, AN/VRC-53, AN/GRC-125, AN-GRC-160, and AN/GRC-163. In addition, the manual band selector switch provided in this antenna permits Antenna AS-1729/VRC to be used with any radio set capable of operating between 30 and 76 MHz provided its output power does not exceed 70 watts.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. If required, the specific issue of individual documents referenced.

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c. Production pack(s) as follows:

Makeup of pack(s)

Number of each kind of pack to be submitted.
Inspection to be performed thereon.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Packaging (see 5.1).

6.3 Color. The color chip furnished by the acquisition agency will be color chip No. 24087, may be obtained upon request to Commanding Officer, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia PA 19120.

6.4 Group C inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.2).

6.5 Location of air-seal test. It is desirable that the air-seal test (see 4.9) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive verification provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality system and the quality history of the product.

6.8 Theory of operation. Antenna AS-1729/VRC is electrically equivalent to a vertical dipole of constant length whose current distribution throughout the operating band is optimized by a pre-selected termination within the base assembly unit. The upper element of the dipole is a straight-forward conductor. The lower element of the dipole is the outside surface of a coaxial cable which is terminated, at the bottom end, against ground in a controlled admittance. The location near the center of the whip, where the upper element joins the inner conductor of the coaxial cable, is referred to as the feed-point. The coaxial cable making up the lower element has two additional functions. It serves to supply rf energy to the feed-point and also serves as a coaxial transformer section grouping the input impedance with a 3:1 VSWR. The termination within the base assembly unit is a ferrite core rf choke whose susceptance to ground is changed (in ten sub-bands) by alternately shunting nine of the sub-bands with present variable components. Sub-band number two is controlled by the base choke alone and is not adjustable. The shunting is accomplished by a step-action rotary solenoid driven switch, which is controlled by a master wafer built into Radio Set AN-VRC-12. A manual override knob serves to indicate the sub-band in operation and provides a capability for manual selection of the desired sub-band when the antenna is used with a radio set other than those listed in 6.1.

6.9 Antenna test chamber. The antenna test chamber is used to facilitate indoor electrical testing of Antenna Matching Unit-Base MX-6707/VRC during production. The chamber is a twelve inch diameter tube, six feet long, the interior contains either ferrite tile or ferrite rings. A modified lower whip section is mounted axially within the chamber. A nine inch copper disk with a short stub at the center that has been tailored to electrically represent the top whip section, screws into the top end of the modified lower whip section. The chamber provides a unique correlation between the impedance of an antenna within the chamber and the same antenna over a reference ten feet by ten feet ground screen in free space. Because the antenna impedance in the chamber at some frequency in each sub-band is within the VSWR requirements of this specification, meaningful rf power tests on the MX-6707/VRC can be accomplished in the chamber.

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6.10 DEFINITIONS

6.10.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.10.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.10.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.10.4 Sporulation. Sporulation is formation of minute unicellular reproductive or dormant bodies, called spores.

6.10.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.11 Matching Unit Base Alignment Cavity. This cavity (see appendix) is a recent development of the U.S. Army Communication - Electronics Command and is considered to be a significant improvement in MX-6707/VRC alignment and test method.

6.12 Subject term (keyword) listing.
Aerial
Vehicular antenna
Whip antenna

6.13 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmentally Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website at http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the list of 31 priority chemicals are cadmium, lead, and mercury. Use of the materials on the list should be minimized or eliminated unless needed to meet the requirements specified herein (see section 3).

6.14 Tin whisker growth (see 3.13.1). The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.15 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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APPENDIX A

CALIBRATION AND ALIGNMENT
PROCEDURES FOR THE MATCHING
UNIT-BASE ALIGNMENT CAVITY

A.1 Scope. This appendix describes the procedures for calibrating the Matching Unit-Base Alignment Cavity and using the Matching Unit-Base Alignment Cavity to align the MX-6707/VRC units during production. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

CECOM DWG SM-B-542023 Alignment Specification

(Copies are available online at http://www.monmouth.army.mil/cecom/ or may be obtained from U. S. Army Communications, Electronics Command, HQ CECOM, AMSEL-PE-OD, Ft. Monmouth, NJ 07703-5000.)

A.3 Description. The Matching Unit-Base Alignment Cavity was developed to facilitate electrical alignment and testing of Antenna Matching Unit-Base MX-6707/VRC during production. The cavity is a rectangular shaped aluminum unit approximately ten inches by seven inches by six inches with a three and one half inch diameter hole in the top for insertion of inverted MX-6707/VRC (less cable-mount assembly SM-D-542042). The cavity contains a step-action rotary solenoid driven switch, adjustable circuit elements (capacitors and/or inductors), and an rf detector circuit.

A.4 Calibration procedure for the Matching Unit-Base Alignment Cavity.

A.4.1 Standard MX-6707/VRC. A standard MX-6707/VRC shall be established in accordance with the following procedure:

a. Select an MX-6707/VRC from preproduction or first production units in which variable shunt elements C8 and L3 through L10 have been aligned on a Wayne-Kerr Admittance Bridge Model B-801-as close as possible to the target values specified in table I of CECOM DWG SM-B-542023.

b. Select a whip assembly (Antenna Elements AS-1095/VRC, AS-1730/VRC and Cable-Mount Assembly SM-D-542042) which meets the susceptance and conductance requirements as specified in 3.8.4.3 and 4.8.4.3 of this specification.

c. Assemble the MX-6707/VRC (aligned in accordance with A.4.1a) and the Whip Assembly (selected in accordance with A.4.1b to form an Antenna AS-1729/VRC.

d. Position the AS-1729/VRC Antenna in an area which meets the standard test conditions delineated in 4.8 of this specification. Figure 3 of this appendix is a block diagram showing a typical free-space test arrangement for measuring the impedance of the antenna.

e. Fine tune variable shunt element C8 and L3 through L10 of MX-6707/VRC at the upper and lower frequencies in each sub-band listed in table III of this specification until the impedance (Z,0) values obtained yield equal VSWR values at the measured frequencies which are within the limits specified in table III of this specification.

f. Remove the AS-1730/VRC, AT-1095/VRC and Cable-Mount Assembly SM-D-542042 from the AS-1729/VRC and perform admittance testing in accordance with CECOM DWG SM-B-542023 on the MX-6707/VRC which has been fine tuned for VSWR on ground plan in accordance with A.4.1.e above to assure that it still meets the admittance values specified in table I of CECOM DWG SM-B-542023. (Table A-I of this specification.)

g. The MX-6707/VRC which has been tested in accordance with A.4.1.a through A.4.1.f above shall be used as Standard MX-6707/VRC for calibration of Matching Unit-Base Alignment Cavity.

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h. Whenever design changes are incorporated into the AS-1729/VRC antenna, the contractor shall perform the procedure in accordance with A.4.1.a through A.4.1.f above on redesigned units and shall use the resulting new Standard MX-6707/VRC to recalibrate the Matching Unit-Base Alignment Cavity.

TABLE A-I. Admittance alignment values. 2/ 3/ 4/ 5/
Extracted from SM-B-542023

Sub band	Frequency MHz	Variable shunt component	Susceptance /2πf pF			Maximum conductance millimhos
			Min	Target	Max	
1 1	30.0 1/ 33.0	C8 ---	+19.4 +20.8	+21.4 +23.9	+22.9 +25.4	.08 .09
2 2	33.0 37.0	None ---	+ 8.6 +10.2	+10.6 +12.2	+12.6 +14.2	.09 .09
3 3	37.0 1/ 42.0	L4 ---	0.0 + 3.2	+ 1.7 + 5.7	+ 5.2 + 6.3	.11 .13
4 4	42.0 47.5 1/	--- L6	- 6.4 - 1.9	- 1.4 + 3.0	+ 1.0 + 5.3	.11 .15
5 5	47.5 1/ 53.0	L5 ---	- 7.5 - 1.0	- 3.6 + 0.3	+ 3.0 + 2.0	.12 .13
6 6	53.0 1/ 56.0	L3 ---	- 8.8 - 8.0	- 2.8 - 1.0	- 1.8 + 0.8	.14 .14
7 7	56.0 60.0 1/	--- L8	-10.0 - 7.4	- 6.0 - 3.0	- 5.0 - 1.8	.14 .16
8 8	60.0 65.0 1/	--- L9	-16.0 -12.0	-10.5 - 7.4	- 9.3 - 6.2	.20 .20
9 9	65.0 70.5 1/	--- L10	-21.4 -17.8	-12.1 - 8.8	-10.5 - 7.8	.20 .25
10 10	70.5 75.95 1/	--- L7	-21.2 -18.1	-14.3 -11.4	-12.8 -10.4	.25 .30

1/ Adjustment frequency.

2/ When aligning base assembly, the variable shunt component shall be adjusted to obtain Susceptance/2πf values as close as possible to above indicated target values.

3/ The Base Alignment Test Fixture SM-B-542114 shall be used in place of Matching Case SM-D-542066 during the calibration of the Matching Unit-Base Alignment Cavity.

4/ For each Matching Unit-Base Cavity, the contractor shall establish and maintain a Standard MX-6707/VRC in accordance with A.4.1 of this appendix.

5/ The contractor maintained Standard MX-6707/VRC unit shall be used at the beginning of each work day, and whenever there is a questionable reading, to verify that the Matching Unit-Base Alignment Cavity is operating satisfactorily.

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A.4.2 Calibration procedures. 1/

a. Set up the equipment as shown in figure A-1.

b. Insert standard MX-6707/VRC (less cable-mount assembly SM-D-542042) into Matching Unit-Base Alignment Cavity and secure with grounding pins.

c. Connect control cable from Alignment Cavity control to connector J2 of standard MX-6707/VRC unit.

d. Turn on equipment and allow to warm up for a minimum of 30 minutes.

e. Set the Alignment Cavity control sub-band selector switch to sub-band 1.

f. Depress 10 MHz marker switch on sweep generator.

g. Depress 10 MHz sweep width switch on sweep generator.

h. Adjust the sweep generator variable sweep width control to maximum (initially).

i. Set the sweep generator sweep mode/time switch to .01 seconds.

j. Set the counter measurement switch to .01 seconds.

k. Adjust the sweep generator marker size control for easily observable marker display on oscilloscope.

l. Adjust the sweep generator variable sweep width control for approximately 4 MHz display bandwidth on oscilloscope with 40 MHz marker at center. Adjust center frequency control as necessary.

m. Adjust the sweep generator variable amplitude control for approximately full vertical screen display on the oscilloscope.

n. Adjust variable component inside Matching Unit-Base Alignment Cavity corresponding to sub-band control switch position to position the 40 MHz marker at the peak of the response curve.

o. Advance the Alignment Cavity control sub-band selector switch to next sub-band.

p. Repeat step n.

q. Repeat steps o and p until all sub-bands have been adjusted.

NOTE:

1/ The Base Alignment Test Fixture SM-B-542114 shall be used in place of Matching Unit Case SM-D-542066 during calibration of the Matching Unit Base Alignment Cavity.

A.5 Production procedure for alignment of MX-6707/VRC Antenna Matching Unit-Base.

a. Set up equipment as shown in figure A-1.

b. Insert untuned production MX-6707/VRC (less cable mount assembly SM-D-542042) into Matching Unit-Base Alignment Cavity and secure with grounding pins.

c. Connect control cable from the Alignment Cavity control connector J2 of MX-6707/VRC.

d. Turn on equipment and allow to warm up for a minimum of 30 minutes.
e. Set the Alignment Cavity Control sub-band selector switch to sub-band 1.

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f. Depress 10 MHz marker switch on sweep generator.

g. Depress 10 MHz sweep width switch on sweep generator.

h. Adjust the sweep generator variable sweep width control to maximum (initially).

i. Set the sweep generator sweep mode/time switch to .01 seconds.

j. Set the counter measurement switch to .01 seconds.

k. Adjust the sweep generator marker size control for easily observed marker display on oscilloscope.

l. Adjust the sweep generator variable sweep width control for approximately 4 MHz display bandwidth on oscilloscope with 40 MHz marker at center. Adjust center frequency control as necessary.

m. Adjust the sweep generator variable amplitude control for approximately full vertical screen display on the oscilloscope.

n. Adjust variable component inside MX-6707/VRC corresponding to sub-band control switch position (see table A-II) to position of the 40 MHz marker at the peak of the response curve.

o. Advance the Alignment Cavity control sub -band selector switch to the next sub-band.

p. Repeat step n.

q. Repeat steps o and p until all sub- bands have been adjusted.

_____

NOTES:

1/ The Base Alignment Test Fixture SM-B-542114 shall be used in place of Matching Unit Case SM-D-542066 during alignment of MX-6707/VRC.

2/ All MX-6707//VRC production units shall be aligned in accordance with the above alignment procedure.

3/ For each Matching Unit-Base Alignment Cavity the contractor shall establish and maintain a Standard MX-6707/VRC in accordance with A.4.1 of this appendix.

4/ The contractor maintained Standard MX-6707/VRC unit shall be used at the beginning of each work day, or whenever there is a questionable reading, to verify that the Matching Unit-Base Alignment Cavity is operating satisfactorily.

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APPENDIX A

TABLE A-II. MX-6707/VRC Variable Components.

Sub - band	Variable component
1	C 8
2	None
3	L 4
4	L 6
5	L 5
6	L 3
7	L 8
8	L 9
9	L 10
10	L 7

A.6. Test equipment required for figure A-1.

Description	Manufacturer and model number
Matching Unit-Base Alignment Cavity	(Contractor Furnished)
Alignment Cavity Control	(Contractor Furnished)
Main Frame	Tektronix TM 504 option 7
Sweep Generator	Tektronix SW 503
Oscilloscope	Tektronix SC 502
Counter	Tektronix DC 502 option 7

NOTE: Unless authorized by the contracting officer, no equivalent to the specified test equipment will be permitted.

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APPENDIX A

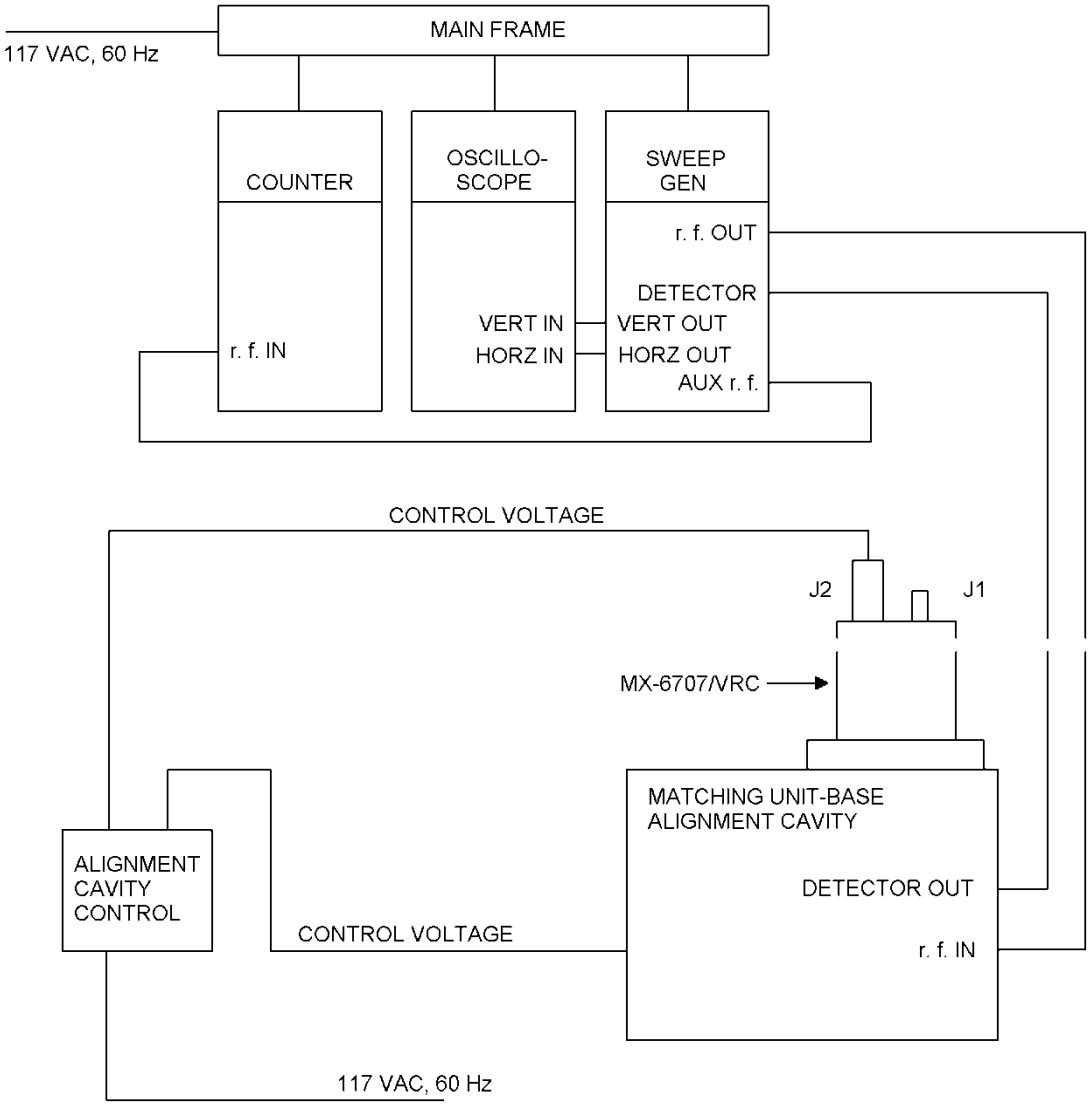


FIGURE A-1. Test setup for use by Matching Unit-Base Alignment Cavity.

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A.7. Test equipment required for figure A-2.

Description	Manufacturer and model number
Vector Impedance Meter	Hewlett-Packard 4815A (H0-1)
Frequency Counter	Hewlett-Packard 5381A
DC Power Supply	Hewlett-Packard 6290A
Band Selector Test Switch	See CECOM DWG SM-B-542023

NOTE: Unless authorized by the contracting officer, no equivalent to the specified test equipment will be permitted.

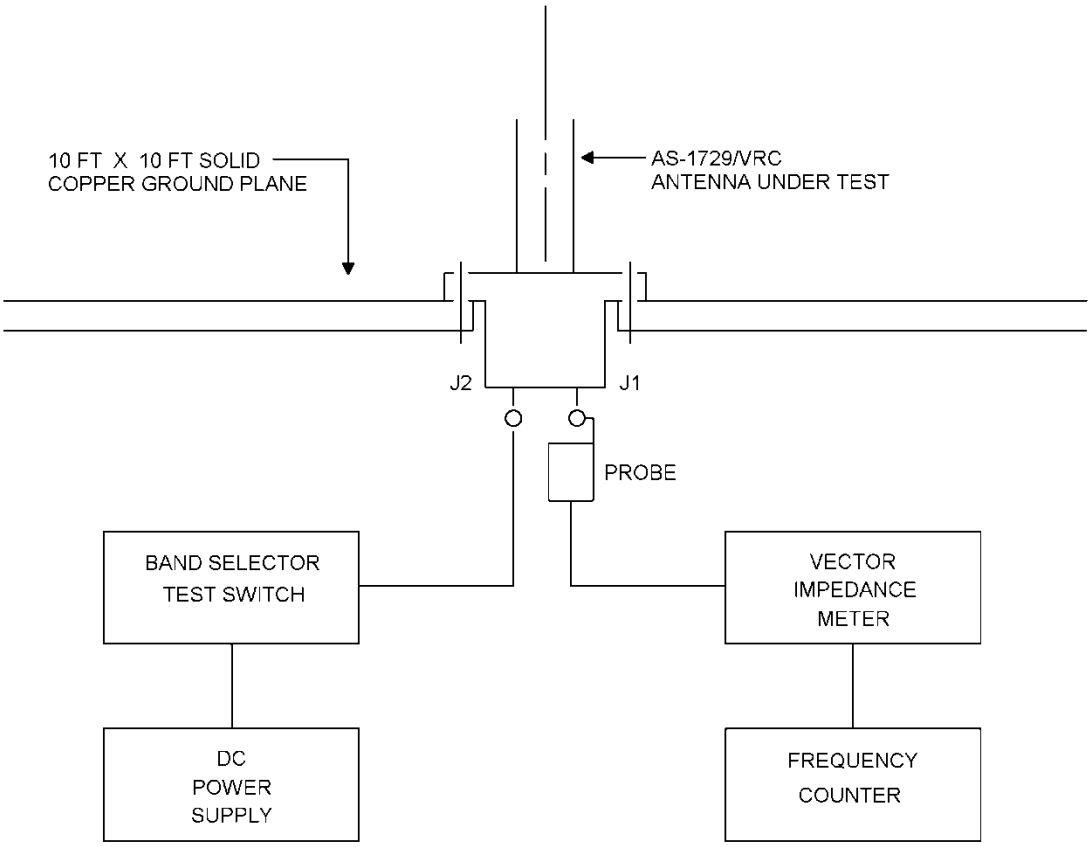


FIGURE A-2. Test setup for measuring impedance of AS-1729/VRC Antenna on ground plane.

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